

Weather API using Python – Detailed Notes

1. Introduction to API

API stands for **Application Programming Interface**.

An API allows two software applications to communicate with each other.

Example:

- Your Python program sends a request to a weather server.
- The server sends weather information back to your program.

So, API acts as a bridge between:

- Client (your Python code)
 - Server (OpenWeatherMap)
-

Real Life Example of API

Suppose you use a weather mobile application.

The app itself does not measure weather.

Instead:

1. App sends request to weather server
2. Server returns weather data
3. App displays temperature, humidity, etc.

This communication happens through APIs.

2. OpenWeatherMap API

OpenWeatherMap provides:

- Current weather data
- 5-day weather forecast
- Air pollution data

- Geographical coordinates
- Historical weather

Website:

[OpenWeatherMap API](#)

3. API Key

To use OpenWeatherMap API:

1. Create account
2. Generate API Key

API key is like a password.

Example:

```
api_key = "YOUR_API_KEY"
```

Without API key:

- API will reject request
-

4. Required Python Libraries

requests

Used for:

- Sending HTTP requests

Install:

```
pip install requests
```

json

Used for:

- Handling JSON data

Already available in Python.

5. What is JSON?

JSON stands for:

JavaScript Object Notation

It is a lightweight data format used in APIs.

Example JSON:

```
{  
  "name": "Ahmedabad",  
  "temp": 32,  
  "humidity": 45  
}
```

6. Geocoding API

Purpose

Converts:

City Name → Latitude & Longitude

Example:

```
Ahmedabad  
→ Latitude = 23.0216238  
→ Longitude = 72.5797068
```

Documentation:

[Geocoding API Documentation](#)

7. Geocoding API Code

```
import requests
import json

def get_coordinates():

    api_key = input('Enter your API key: ')
    city = input('Enter the city name: ')

    request_url = (
        "http://api.openweathermap.org/geo/1.0/direct?q="
        + city
        + "&limit=1&appid="
        + api_key
    )

    response = requests.get(request_url)

    coordinates = response.json()

    print(json.dumps(coordinates, indent=5))

    latitude = coordinates[0]['lat']
    longitude = coordinates[0]['lon']

    return (latitude, longitude)

lat, long = get_coordinates()

print("Latitude :", lat)
print("Longitude :", long)
```

8. Explanation of Geocoding Code

Step 1 – Import Libraries

```
import requests
import json
```

- requests → API communication
 - json → JSON formatting
-

Step 2 – Function Creation

```
def get_coordinates():
```

Creates a function.

Step 3 – User Input

```
api_key = input()
city = input()
```

Takes:

- API key
 - City name
-

Step 4 – Create URL

```
request_url = ...
```

Creates API request URL.

Example:

```
http://api.openweathermap.org/geo/1.0/direct?q=Ahmedabad&limit=1&appid=API_KEY
```

Step 5 – Send Request

```
response = requests.get(request_url)
```

Sends request to API server.

Step 6 – Convert JSON

```
coordinates = response.json()
```

Converts response into Python object.

Step 7 – Print JSON

```
print(json.dumps(coordinates, indent=5))
```

Prints formatted JSON.

Step 8 – Extract Data

```
latitude = coordinates[0]['lat']  
longitude = coordinates[0]['lon']
```

Gets latitude and longitude.

9. Current Weather API

Purpose

Provides:

- Temperature
- Pressure
- Humidity
- Wind speed
- Visibility
- Weather condition

Documentation:

[Current Weather API](#)

10. Current Weather API Code

```
import requests  
import json  
  
def display_current_weather():  
    api_key = input('Enter your API key: ')  
    request_url = (  
        "https://api.openweathermap.org/data/2.5/weather?lat=" +  
        str(lat)
```

```

        + "&lon="
        + str(long)
        + "&appid="
        + api_key
        + "&units=metric"
    )

    response = requests.get(request_url)

    current_weather = response.json()

    print(json.dumps(current_weather, indent=5))

    description = current_weather['weather'][0]['description']
    temperature = current_weather['main']['temp']
    pressure = current_weather['main']['pressure']
    humidity = current_weather['main']['humidity']
    wind_speed = current_weather['wind']['speed']
    visibility = current_weather['visibility']

    print('Current Weather')
    print('Temperature', temperature)
    print('Pressure', pressure)
    print('Humidity', humidity)
    print('Wind Speed', wind_speed)
    print('Visibility', visibility)
    print('Description', description)

display_current_weather()

```

11. Weather Information Explanation

Field	Meaning
temp	Temperature
humidity	Moisture in air
pressure	Atmospheric pressure
wind_speed	Wind speed
visibility	Visibility distance
description	Weather condition

12. Air Pollution API

Purpose

Provides:

- Air Quality Index (AQI)
- Pollutant levels

Documentation:

[Air Pollution API](#)

13. Air Pollution API Code

```
import requests
import json

quality_indices = {
    1: 'Good',
    2: 'Fair',
    3: 'Moderate',
    4: 'Poor',
    5: 'Very Poor'
}

def display_current_air_pollution():

    api_key = input('Enter your API key: ')

    request_url = (
        "http://api.openweathermap.org/data/2.5/air_pollution?lat="
        + str(lat)
        + "&lon="
        + str(long)
        + "&appid="
        + api_key
    )

    response = requests.get(request_url)

    current_pollution = response.json()

    print(json.dumps(current_pollution, indent=5))

    index = current_pollution['list'][0]['main']['aqi']

    print('Current Air Quality :', quality_indices[index])

display_current_air_pollution()
```

14. AQI Meaning

AQI Meaning

- 1 Good
- 2 Fair
- 3 Moderate
- 4 Poor
- 5 Very Poor